

In-Situ Observations of the Marquesas Island Mass Effect

Chloe Cheng, Yuchen Li

Stanford University

SEA Cruise S315

May 9—June 12 2024

1 Introduction

The Marquesas Islands form an archipelago in the central South Pacific with surrounding waters featuring persistent enhancement of surface chlorophyll relative to the oligotrophic open ocean. Increased primary productivity in island wakes is a rather ubiquitous phenomenon known as the Island Mass Effect (De Falco et al. 2022; Doty and Oguri 1956, IME).

The particular mechanisms driving the Marquesas IME are not yet clear. It occurs in a high nutrient-low chlorophyll (HNLC) region, where macronutrients such as nitrate are typically sufficient for phytoplankton growth but primary productivity is limited (Martinez and Maamaatuaiahutapu 2004; Signorini et al. 1999). In this case, the limiting nutrient is thought to be the micronutrient iron, where the concentration near the Marquesas was measured to be between 0.15 and 0.42 nmol (Blain et al. 2008).

Previous hypotheses regarding the Marquesas IME can be broadly grouped into three main categories: (1) subsurface iron is introduced to the mixed layer away from island masses (Martinez and Maamaatuaiahutapu 2004; Raapoto et al. 2018), (2) iron is introduced to waters from the islands (Raapoto et al. 2019; Signorini et al. 1999), and (3) upwelling of the deep chlorophyll maximum (DCM) creates an apparent phytoplankton bloom that does not involve new growth of phytoplankton (Hasegawa et al. 2009, 2008). We summarize these existing hypotheses below.

Many explanations for the Marquesas IME involve eddy-driven upwelling. Islands act as an obstruction to surface currents that, depending on current velocity and island geometry can lead to the formation of downstream vortices. Cyclonic eddies may bring sub-pycnocline iron up to the mixed layer and then produce phytoplankton blooms. Raapoto et al. (2018) used a 4-km resolution configuration of the Regional Ocean Modeling System (ROMS) to simulate vertical motion induced by eddies in the wake of the Marquesas. They indeed find significant vertical velocities associated with flow separation.

Another set of hypotheses suggest that the islands themselves can be a source of dissolved iron. For example, iron can be dissolved in runoff and discharged directly into the ocean or as iron-rich sediments can be deposited on the island margins that are subsequently leached. However, Martinez and Maamaatuaiahutapu (2004) find no significant correlation between rainfall events and peaks of chlorophyll, which suggests that runoff may not be sufficient to initiate a bloom. On the contrary, Raapoto et al. (2019) run a coupled physical-biogeochemical regional model and find that iron introduced from island runoff reproduces

somewhat realistic chlorophyll variability, though phytoplankton blooms in their simulations are too weak compared to observations.

The last hypothesis posits that apparent blooms at the surface consist of upwelled existing subsurface phytoplankton as opposed to new production. Evidence for this hypothesis could come from subsurface observations of how the DCM is modified by vertical mixing. There is a Biogeochemical Argo (BGC-Argo) profiling float (Wong et al. 2020) that was deployed west of Nuku Hiva, the largest island in the archipelago. Raapoto et al. (2019) noted variation in Chl-a profiles, with some featuring a distinct DCM and others showing mixing in the upper ocean.

Here we add to the currently sparse collection of existing in-situ data using the oceanographic research sailing vessel *Robert C. Seamans* operated by the Sea Education Association in late May 2024. This work aims to use further in-situ observations of subsurface chlorophyll and water column structure, in conjunction with remotely sensed data, to provide better understanding of the drivers of the IME specific to this region, which may also serve as clues to other IMEs globally. Building on previous knowledge of near-island mechanisms for the IME, we suspect that there are multiple processes at play that lead to enhanced productivity.

2 Data & Methods

During our cruise, we conduct six transects using a mini CTD (RBR *Concerto*) towed in an alternating up and down motion. The CTD is also equipped with a fluorometer to measure chlorophyll *a* concentration. The locations of these transects are picked to sample along gradients of chlorophyll *a* concentration, thereby sampling the water column both inside and outside a region affected by IME processes. Since the towing winch on the ship can only be operated safely while on the high side, our transects were limited to ones that oriented the ship such that wind came from the port side. The depth of the tows was chosen to capture both the subsurface chlorophyll maximum (which in this region is typically at 50 meters) as well as the pycnocline (typically 120 meters).

The fluorometer is calibrated using a chlorophyll extraction procedure conducted over 45 Niskin bottle samples from our twice-daily CTD hydrocasts. An additional calibration step was required due to differences between the CTD on the rosette and the mini CTD that was towed for the transects (see Appendix A for details).

We also use the shipboard acoustic doppler current profiler (ADCP) to provide velocity at

various depths in the water column. The shallowest depth bin is 18.7 meters. We use the velocity data to reveal localized flow in the lee of the islands, as well as characterize general characteristics of the flow regime by calculating the Reynolds number Re .

To complement our ship sampled data, we use (1) concurrent satellite products to provide a synoptic picture of the surrounding waters during our transects as well as (2) historical satellite products to study longer timescales of variability. For the cruise-concurrent data, we use surface chlorophyll derived from ocean color (Aqua Modis 3-day product), sea surface temperature, and a blended sea surface height. For historical chlorophyll, we use the weekly GlobColour product, which is derived from multiple ocean color sources extending back to 1998. For historical SST, we use the NOAA OISSTv2 (Optimal Interpolation SST version 2). For historical surface currents, we use the OSCAR dataset (5-day frequency, $1/3^\circ$ horizontal resolution). Finally, for rainfall we use a combination of weather station data from Nuku Hiva and Hiva Oa, integrated to a monthly frequency¹.

3 Results

3.1 In-situ transects

We conducted a total of six towed CTD transects during our cruise, with their locations selected in an attempt to sample the spatial variability of the phytoplankton blooms and their associated physical contexts. The transects are 15 to 20 km in length and reach a maximum depth of 150 meters. We exclude one transect from our analysis that was cut short due to technical difficulties.

The surface chlorophyll concentration during our passage is plotted with the transects, labeled A through E, in Figure 1a,b. A clear enhancement of surface chlorophyll can be seen downstream of the islands that is indicative of an active IME. During May 18-19, we successfully intercepted localized areas of enhanced surface chlorophyll in our A and B transects, exiting one area and entering another, respectively.

In all transects, the DCM outside the highest surface chlorophyll occurs at around 50 meters (Figure 2). For the transects where we intersected a significant gradient in surface chlorophyll, we find that surface chlorophyll increases concurrently with a rise in the DCM to about 30 meters in transect B and up to the surface in transect A. This could reflect either

¹We note that monthly data is not ideal for a phenomenon (namely, runoff-associated phytoplankton blooms) that occurs over a timescale of days to weeks, but this data source was the only one we found while still at shore

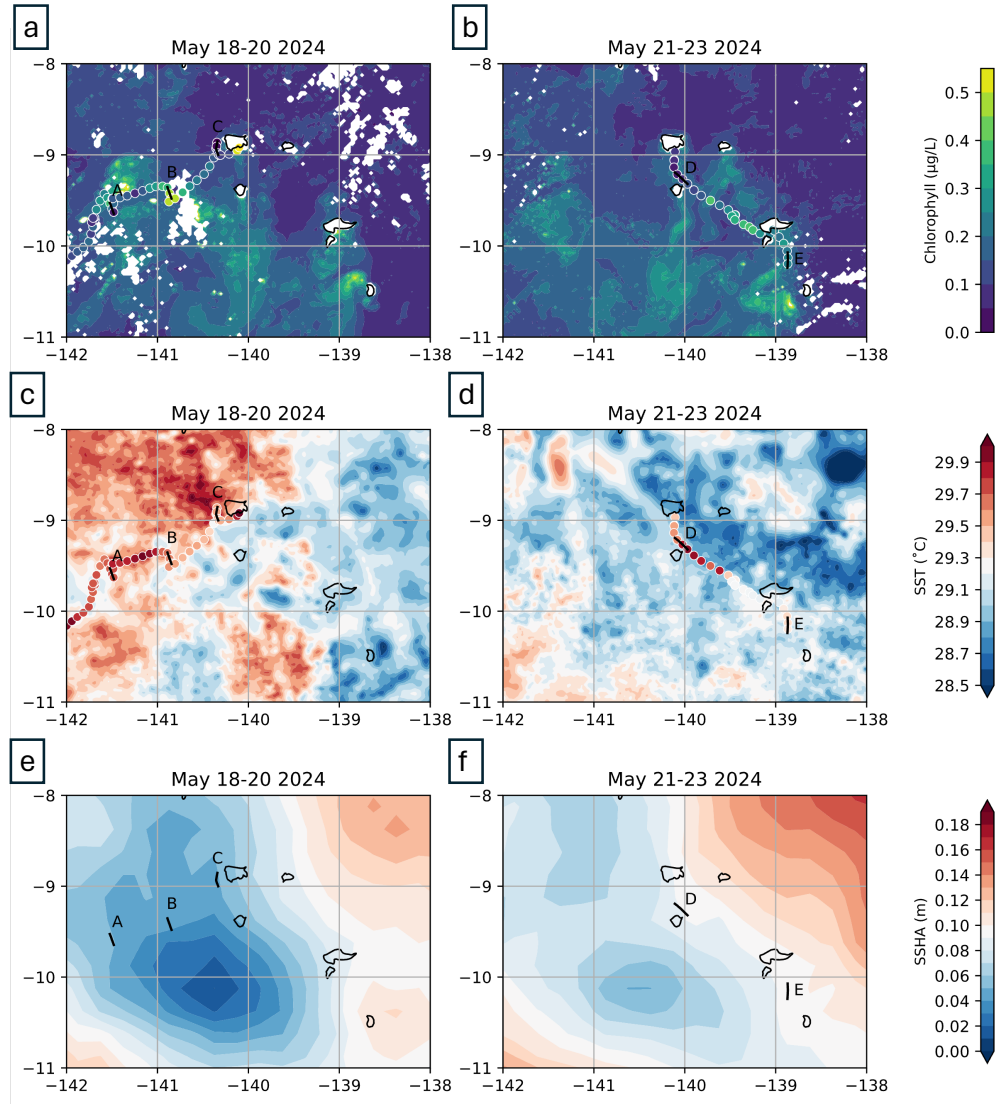


Figure 1: (a) 3-day average surface chlorophyll with surface fluorescence from shipboard flow-through system (dots) from May 18 through 20. Transects A, B, and C on the approach to Nuku Hiva are plotted; (b) Same as (a), but from May 21 through 23. Transects D and E on the departure from Nuku Hiva are plotted; (c) Same as (a), but with SST; (d) Same as (b), but with SST; (e) Same as (a), but with sea surface height anomaly (SSHA); (f) Same as (b), but with SSHA.

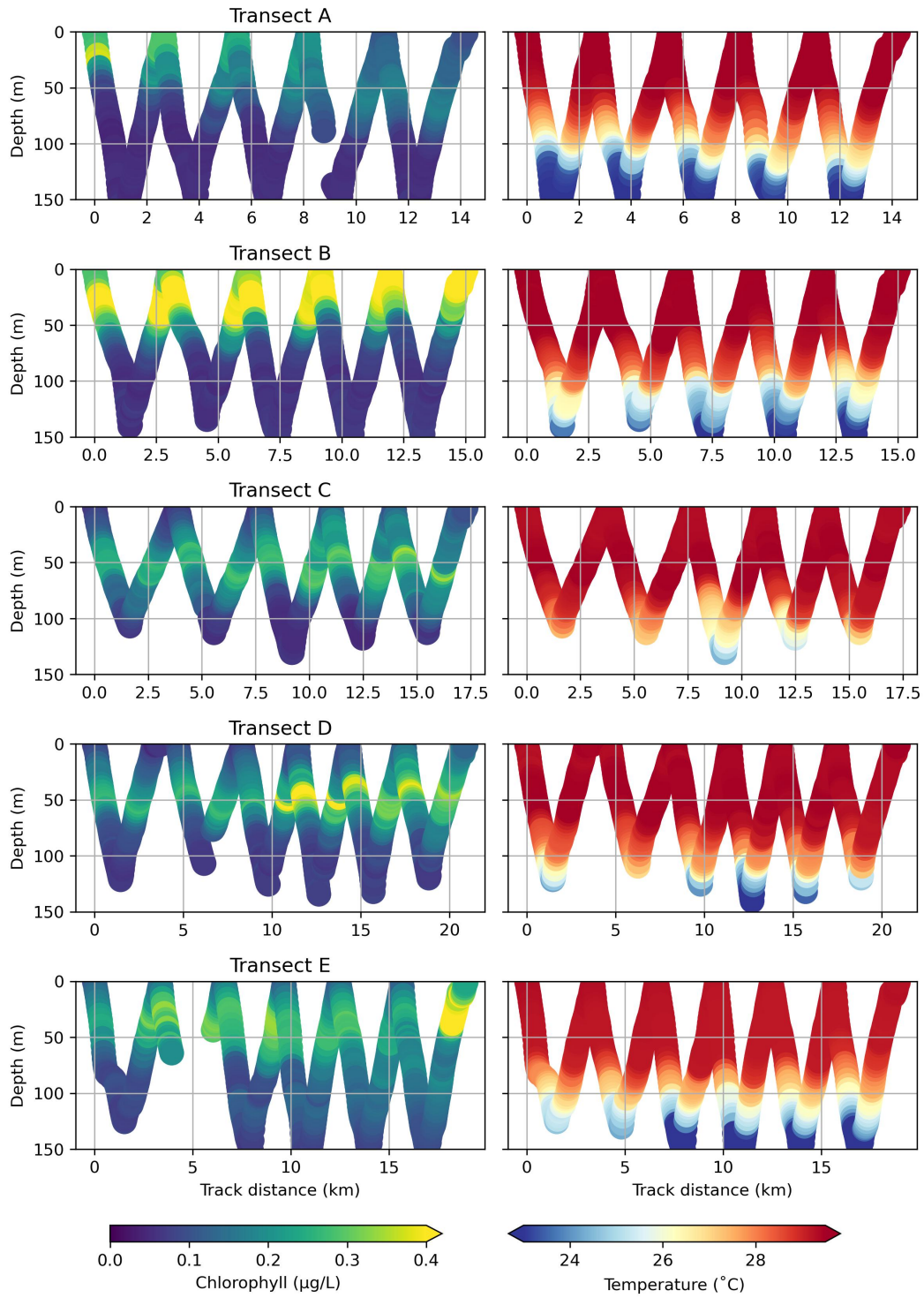


Figure 2: Subsurface profiles of chlorophyll (left) and temperature (right) obtained from CTD tow transects A through E.

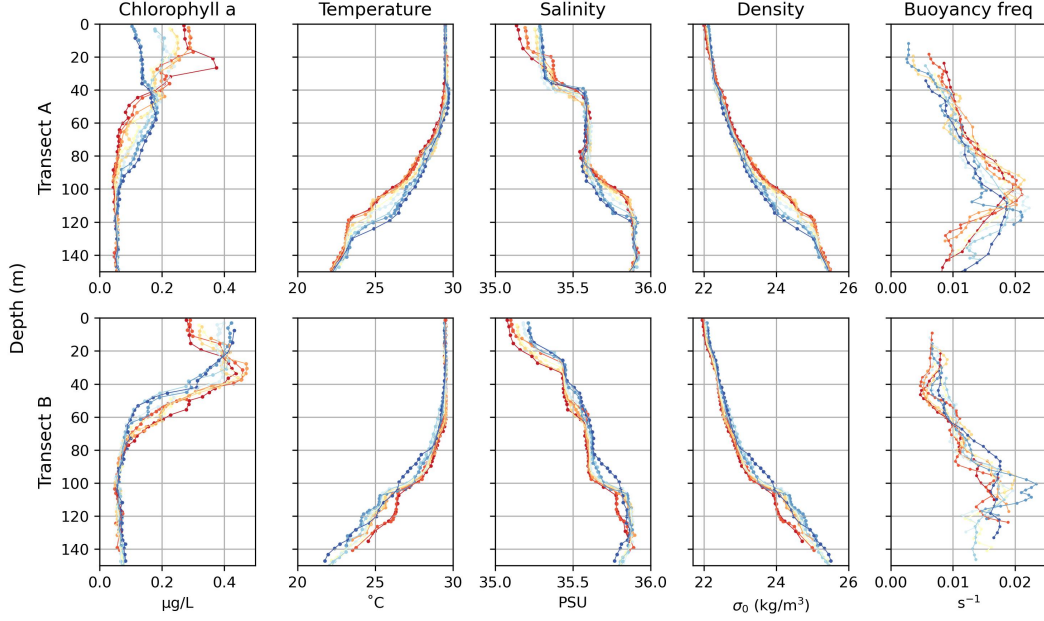


Figure 3: Pseudo-vertical profiles of chlorophyll, temperature, salinity, density, and buoyancy (Brunt-Väisälä) frequency for transects A and B. The color of the profile indicates distance along the transect the profile, with red at the beginning of the transect and blue at the end. The buoyancy frequency profiles are smoothed with a moving mean window of size 9.

a change in the vertical distribution phytoplankton biomass due to differences in the water column structure (see, e.g., Hasegawa et al. 2008) or a change in the vertical distribution of chlorophyll due to differences in light level and photoadaptation.

One indication towards the former hypothesis (but does not necessarily rule out the latter) is that zones with elevated DCM and surface chlorophyll also display a local shoaling of both the pycnocline and thermocline. We observe in the density profile for transect A (Figure 3) that the depth of sharpest vertical density change drops by about 20 meters as we leave the bloom. We confirm this by calculating the buoyancy (Brunt-Väisälä) frequency N , defined as

$$N = \sqrt{-\frac{g}{\rho_0} \frac{\partial \rho}{\partial z}}$$

where ρ is the in-situ density and ρ_0 is a reference density. It is proportional to the stratification of the water column. Indeed, the peak in buoyancy frequency (thus, where the vertical density gradient reaches its maximum) lowers by the same amount (rightmost subplots in Figure 3). Transect B shows similar trends but in reverse, since we leave the bloom during that transect. In summary, it appears that there is a lifting of the pycnocline, seen as a

relatively stable layer in the water column, that also lifts the DCM that sits above it.

Generally in many cases, a signature of upwelling can be seen at the surface as a region of cooler SSTs. Interestingly, significant changes in the thermal and density structure of the water column occur only at depths of 75 to 120 meters (Figure 3).

We also observe a change in chlorophyll concentration at the DCM in the A and B transects with changes in surface chlorophyll. The shallower DCM within blooms occurs at greater concentrations than the deeper DCM further from the higher surface chlorophyll. This is particularly marked in transect A.

Transects C-E crossed much weaker surface chlorophyll gradients and likely missed the strongest surface chlorophyll blooms. They exhibit relatively low surface chlorophyll concentrations (less than $0.1 \mu\text{g/L}$). Profiles as shown in Figure 3 are provided for these transects as supplementary figures.

3.2 Surface maps

To establish the regional context of the phytoplankton blooms observed during our cruise, we show satellite products of surface chlorophyll (Figure 1a,b), SST (1c,d), and sea surface height anomaly (1e,f). Zones of enhanced surface chlorophyll can be seen emerging in the lee of the islands with a south to southwest trend.

As noted in the previous section, we find little change in SST as we entered a region of higher surface chlorophyll. There are several patches of eddy-like SST variability (Figure 1c,d) on the same length scale as the islands (and therefore possibly eddies generated by instability in the island wake); however, without more detailed dynamical information, it is unclear whether these are truly signatures of eddies. Additionally, the discrepancy between the satellite-derived temperature and that from the ship flow-through system is due to the ship-based measurements being taken from beneath the surface.

Finally, the sea surface height data (Figure 1e,f) is of too low resolution to resolve eddy features on the order of 20-30 km and underscores the need for higher resolution altimetry to better understand mesoscale to submesoscale structures in the ocean. The new Surface Water and Ocean Topography (SWOT) satellite has the potential to fill this gap (Callies and Wu 2019) and provides a potential opportunity for future research on eddy features in island lees.

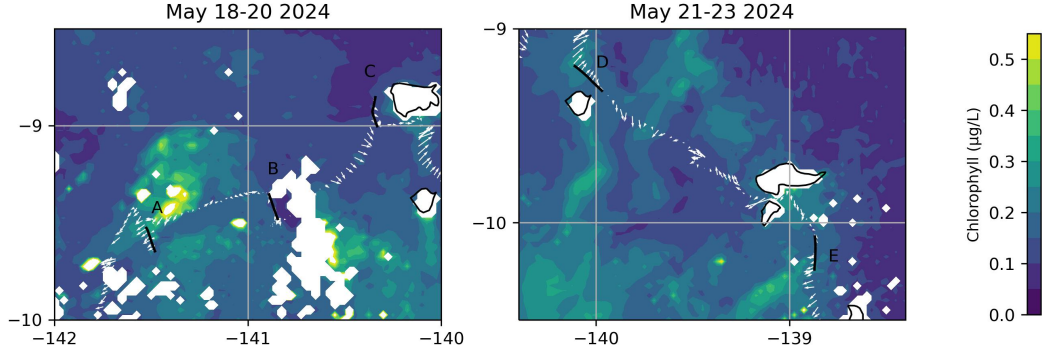


Figure 4: Surface currents (arrows) superimposed on surface chlorophyll from May 18 to May 20 (left) and from May 21 to May 23 (right).

3.3 Physical mechanisms and dynamical regime

Our shipboard ADCP yields horizontal current velocities only. Currents measured from the shallowest depth of 18 meters are different from those in next tens of meters, so we plot currents from 28 meters along our cruise track.

Just to the northeast of transect A is the greatest concentration in the high chlorophyll region, where we also see currents appearing to diverge. This is less apparent around transect B, where currents are relatively weak overall. These changes in currents are unlikely due solely to current changes over time as we traversed each transect in no more than 3 hours. Note that the surface chlorophyll in Figure 4 exhibits some discrepancies with chlorophyll measurements from the ship flow-through system; we observed greater changes in surface chlorophyll concentration along transects A and B than visible in this figure.

Diverging currents and very localized high chlorophyll regions seem to indicate the presence of eddies. However, we cannot determine directly how much, if at all, these potential eddies lead to upwelling. We attempted to use our track velocity data to estimate vertical velocities under the assumption that the 1-dimensional divergence in the direction of the track approximates the actual 2-dimensional divergence. This does not seem to hold, so upwelling cannot be derived from current data alone.

Previous studies of island wakes have noted the usefulness of the nondimensional Reynolds number Re , which arises naturally as a characteristic scaling parameter in systems where a viscous fluid flows around an obstacle (Coutis and Middleton 1999; De Falco et al. 2022; Hasegawa et al. 2009; Heywood et al. 1990). It is defined as the ratio of inertial to viscous

forces:

$$Re = \frac{UL}{\nu}$$

where U is a characteristic velocity scale, L is a characteristic length scale, and ν is the viscosity of the fluid. Here we estimate the Reynolds number using surface current velocities obtained from the ADCP. We take the length scale to be the diameter of the island obstacle (30 km for Nuku Hiva), and calculate the average flow speed from our velocity data within a certain distance from Nuku Hiva. We use a standard value of $\nu = 100 \text{ m}^2/\text{s}$ for the viscosity parameter (Raapoto et al. 2018). The calculated value of Re is sensitive to how much of the ship track is included in the bounding box used for averaging, so we use a bootstrapping approach over ranges of bounds. We get a median Reynolds number of 33, which is consistent with a qualitative description of the flow regime: far from laminar yet not featuring a vortex street. Typically $Re > 40$ is required for the formation of Von Kármán vortex streets, in which case we would expect to see coherent and well-defined eddies forming far (e.g., extending 100 km) from the island masses.

During the cruise, it is reasonable to predict a lower Re value because surface currents in the Marquesas were relatively weak. Overall currents differ greatly from their prevailing southeasterly direction and have an average speed of about 0.1 m/s (Figure 4). That eddies may be present, but that they likely did not form far from islands in a vortex street, suggests that the eddies intercepted by A and B had formed in the immediate wake lee of islands and subsequently were advected downstream. Additional evidence may lay in the absence of SST changes across these eddies. As mentioned before, we do not observe the cooling SSTs typically associated with upwelling but see changes in thermocline depth. We interpret these observations to be a sign of past upwelling, where SSTs had equilibrated before we arrived.

4 Discussion

4.1 Seasonal and interannual variability

Due to the inherent limitation in spatial and temporal scope of our transects, we use historical satellite and climate data to study the time variation of the IME over the past 25 years. This is a natural continuation of Martinez and Maamaatuaiahutapu (2004) (hereafter MM04), who analyzed the same time series two decades ago. We are not aware of any publications that have repeated this timeseries analysis since then.

The surface chlorophyll concentration displays a clear seasonal cycle (Figure 5, 6a, S3),

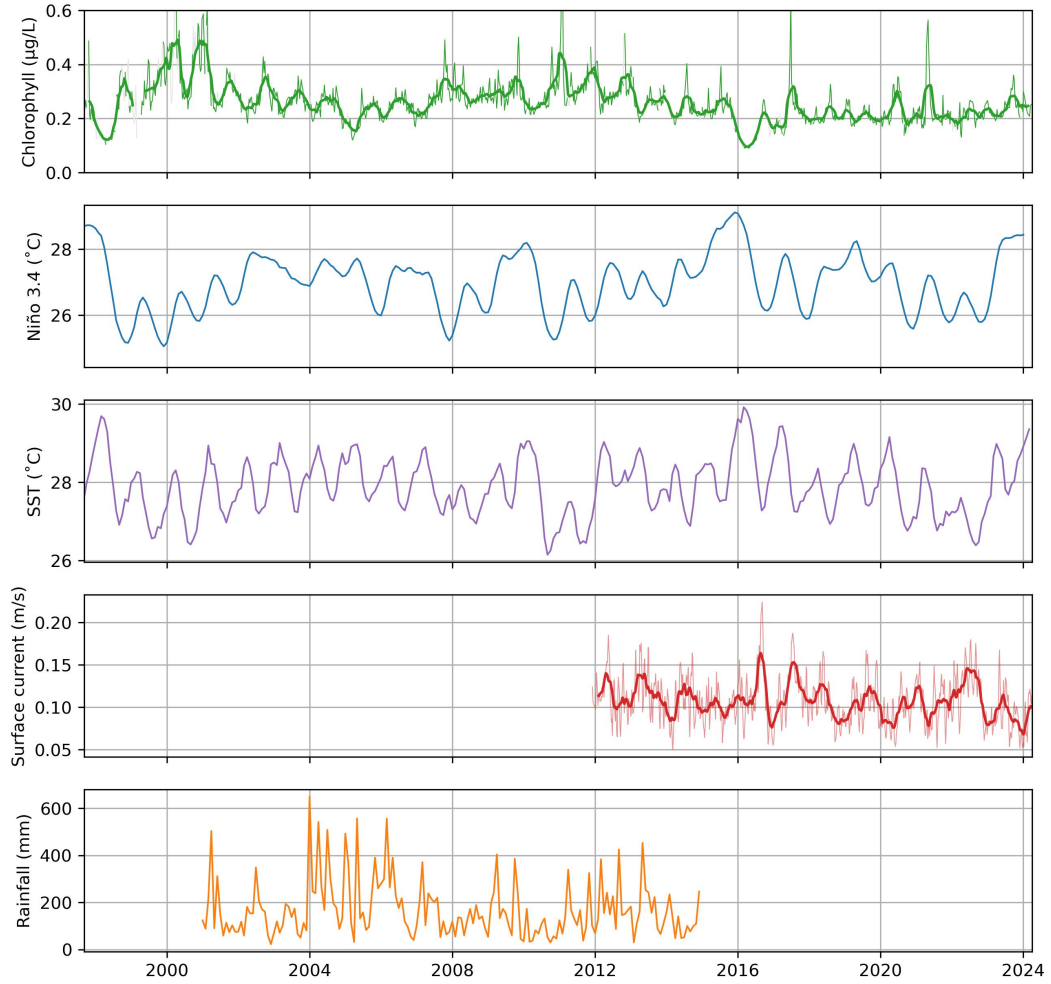


Figure 5: (a): Weekly surface chlorophyll concentration averaged over $8\text{--}11^{\circ}\text{ S}$ and $138\text{--}142^{\circ}\text{ W}$. The bold line is a moving average with a width of 3 months, therefore emphasizing seasonal variability; (b): Niño 3.4 ENSO SST index; (c) Monthly SST averaged over the same region as in (a); (d): 5-day surface current speed averaged over the same region as in (a). The bold line is a moving average with a width of 3 months, therefore emphasizing seasonal variability; (e): monthly rainfall from two weather stations on Nuku Hiva and Hiva Oa.

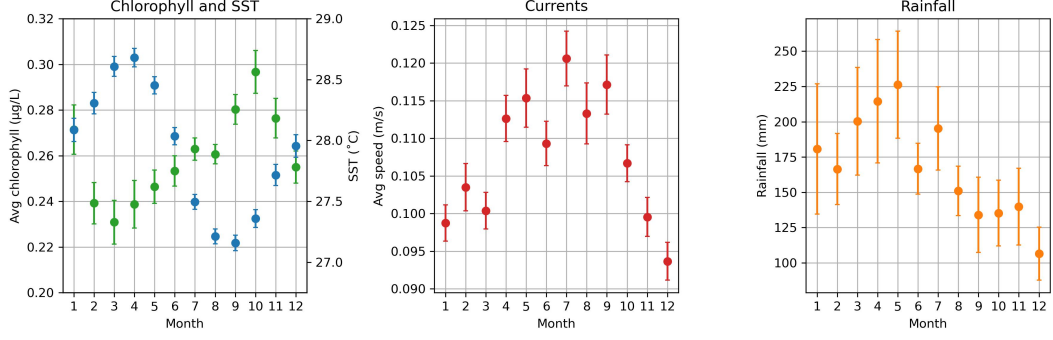


Figure 6: (a): Surface chlorophyll (green) and SST (blue) over the same averaging region as in Figure 5. Error bars show one standard error from the mean; (b): Same as (a), but with currents; (c): Same as (a), but with rainfall.

reaching a maximum in austral spring and reaching a minimum in austral autumn. Raapoto et al. (2019) claim that this seasonal cycle corresponds to the seasonality of wind stress, which is also highly correlated with the eddy kinetic energy in their regional model. Signorini et al. (1999) and later MM04 show that there is a relationship between average speed of surface currents and chlorophyll blooms. They hypothesize that this is due to the dynamical relationship between shear-driven mixing due to island-flow interaction and wind stress driven mixing; as they note, these mechanisms are not separate because greater wind stress is expected to increase the mean surface flow. We repeat the regression between chlorophyll and total surface currents done in MM04. Unlike their analysis, we find no significant relationship between surface current speed and chlorophyll on both weekly ($p = 0.66$) and seasonal timescales ($p = 0.75$). This suggests that the relationship between average surface currents and chlorophyll is not as robust as previously thought. Moreover, it could indicate that the average surface speed is not a good indicator for dynamical flow interactions with the island. The eddy kinetic energy (EKE) is a notable alternative and provides insight into flow variability. Raapoto et al. (2018) ran a high resolution regional model to calculate the EKE seasonal cycle. However, as they show, observational products of surface currents such as OSCAR (which we use here too) are of too low resolution to fully resolve wake eddies. Therefore, the EKE in their model was greater than the EKE derived from OSCAR.

We revisit the runoff hypothesis as in MM04 and Raapoto et al. (2019) by relating the surface chlorophyll to rainfall over the two largest islands of Nuku Hiva and Hiva Oa (Figure 5e). We find a weak but statistically significant ($r = -0.201, p < 0.01$) anticorrelation between the two timeseries. The seasonal cycle of rainfall is shown in Figure 6c, with a maximum in austral autumn and minimum in austral summer. Our results are consistent

with MM04, who found a seasonal cycle of rainfall with a peak in March (therefore opposite to the seasonal chlorophyll cycle). One caveat is that we use monthly rainfall data, which would only provide an time-integrated signal for a phenomenon (runoff-induced blooms) that takes place over 1-10 days. However, if there is a causal relationship between runoff and chlorophyll blooms, one would expect the highest rainfall months to also correspond to the greatest number of blooms. Inspection of weekly surface chlorophyll snapshots during the highest rainfall months do not indicate such a relationship.

Lastly, we relate the surface chlorophyll to both local (Figure 5c) and Equatorial Pacific SST (Figure 5b). On seasonal timescales, chlorophyll is anticorrelated ($r = -0.341, p < 0.001$) with local SST, with the local SST having a pronounced seasonal cycle peaking in March (Figure 6a). One would expect cooler SSTs as a signature for upwelling in the wake. However, this seasonal cycle is most likely dominated by changes in solar insolation, rather than a thermal signature of dynamical changes associated with upwelling and enhanced chlorophyll. Moreover, as noted by Raapoto et al. (2018), there are in fact zones of warmer SST in the immediate lee of the islands (their Figure 3a). This phenomenon has also been observed in other islands (De Falco et al. 2022) and is opposite of what one expects from locally upwelled water. Currently, it is believed to be due to a persistently shallower mixed layer in the immediate lee—which would be warmer given the same thermal forcing, by a volume argument—compared to surrounding waters. We observed no significant change in the mixed layer depth in Transect C (Appendix B), which was closest to the island, compared to other transects. However, the ADCP current data reveals that Transect C was probably not within the wake of the island during our cruise, so we do not have conclusive evidence against this hypothesis.

On interannual timescales, the chlorophyll in the Marquesas is anticorrelated with the Niño 3.4 index ($r = -0.389, p < 0.001$), with higher chlorophyll during La Niña years and lower chlorophyll during El Niño years. This is potentially due to changes in the mixed layer biogeochemistry affected by changes in equatorial upwelling. During El Niño, it is known that parts of the Equatorial Pacific become nitrate-limited (Strutton and Chavez 2000). The propagation of this nitrate-limited water to the Marquesas can take place via the South Equatorial Current and further limit phytoplankton growth. Another possible mechanism is dynamical in origin: the speed of equatorial currents is lower during El Niño years. However, as previously noted, the relationship between current speed and chlorophyll is weak and apparently not stationary. An analysis of ENSO-modulated changes in biogeochemistry and regional circulation is beyond the scope of this paper, but presents itself as a ripe opportunity for future work.

5 Conclusion

We provide a view of the island mass effect in the Marquesas through targeted measurements of the subsurface during a period of average enhanced productivity. From our ship-collected and satellite data, we find that within some of the strongest blooms present during our cruise are shallower DCMs and elevated thermoclines, but no temperature signatures of upwelling at the surface. With eddy-like currents measured near these blooms, we interpret such blooms to be centers of eddy-driven upwelling that likely had occurred in the immediate lee of the islands.

Although not included within our study, determining most directly where and when phytoplankton are actively producing chlorophyll would likely involve characterizing the biogeochemistry of the surrounding waters by sampling for dissolved iron. The concentration of iron in the surface water of the blooms we observed would inform us whether or not there is still upwelling supplying the mixed layer with sub-pycnocline iron. Along with rainfall data, directly sampling for iron off the shore of Nuku Hiva would provide insight as to how much iron is introduced to the mixed layer by island runoff.

We also emphasize the need for further in-situ data over longer timespans. Our limited temporal coverage had allowed us to see the structure of a few select blooms, but without revisiting this region over seasonal to interannual timescales, it is difficult to describe other forms chlorophyll enhancement may take. There is ample opportunity to continue to characterize enhanced chlorophyll around the Marquesas, which can be used, for example, to estimate how much primary production is a result of the advected eddies we observed here.

6 Acknowledgements

We thank the entire SEA staff and Stanford faculty that made this cruise possible. In particular, we thank Jan Witting, Jordan Eckstein, Matt Birhle, Audrey Bennett, Barbara Block, Rob Dunbar, Captain Rick Miller, the TAs, and the 19 other students of Stanford at SEA 2024 for helping us collect our transect data and for providing scientific and moral support.

References

- Blain, S., S. Bonnet, and C. Guieu (Feb. 2008). “Dissolved iron distribution in the tropical and sub tropical South Eastern Pacific”. English. In: *Biogeosciences* 5.1. Publisher: Copernicus GmbH, pp. 269–280. ISSN: 1726-4170. DOI: [10.5194/bg-5-269-2008](https://doi.org/10.5194/bg-5-269-2008). URL: <https://bg.copernicus.org/articles/5/269/2008/> (visited on 04/12/2024).
- Callies, J. and W. Wu (Sept. 2019). “Some Expectations for Submesoscale Sea Surface Height Variance Spectra”. EN. In: *Journal of Physical Oceanography* 49.9. Publisher: American Meteorological Society Section: Journal of Physical Oceanography, pp. 2271–2289. ISSN: 0022-3670, 1520-0485. DOI: [10.1175/JPO-D-18-0272.1](https://doi.org/10.1175/JPO-D-18-0272.1). URL: <https://journals.ametsoc.org/view/journals/phoc/49/9/jpo-d-18-0272.1.xml> (visited on 04/19/2024).
- Coutis, P. F. and J. H. Middleton (Sept. 1999). “Flow-topography interaction in the vicinity of an isolated, deep ocean island”. In: *Deep Sea Research Part I: Oceanographic Research Papers* 46.9, pp. 1633–1652. ISSN: 0967-0637. DOI: [10.1016/S0967-0637\(99\)00007-2](https://doi.org/10.1016/S0967-0637(99)00007-2). URL: <https://www.sciencedirect.com/science/article/pii/S0967063799000072> (visited on 04/19/2024).
- De Falco, C. et al. (July 2022). “Island Mass Effect: A Review of Oceanic Physical Processes”. English. In: *Frontiers in Marine Science* 9. Publisher: Frontiers. ISSN: 2296-7745. DOI: [10.3389/fmars.2022.894860](https://doi.org/10.3389/fmars.2022.894860). URL: <https://www.frontiersin.org/articles/10.3389/fmars.2022.894860> (visited on 04/03/2024).
- Doty, M. S. and M. Oguri (Oct. 1956). “The Island Mass Effect”. In: *ICES Journal of Marine Science* 22.1, pp. 33–37. ISSN: 1054-3139. DOI: [10.1093/icesjms/22.1.33](https://doi.org/10.1093/icesjms/22.1.33). URL: <https://doi.org/10.1093/icesjms/22.1.33> (visited on 04/28/2024).
- Hasegawa, D., M. R. Lewis, and A. Gangopadhyay (2009). “How islands cause phytoplankton to bloom in their wakes”. en. In: *Geophysical Research Letters* 36.20. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/2009GL039743>. ISSN: 1944-8007. DOI: [10.1029/2009GL039743](https://doi.org/10.1029/2009GL039743). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1029/2009GL039743> (visited on 04/03/2024).
- Hasegawa, D. et al. (Feb. 2008). “Apparent phytoplankton bloom due to island mass effect”. In: *Journal of Marine Systems. Physical-Biological Interactions in the Upper Ocean* 69.3, pp. 238–246. ISSN: 0924-7963. DOI: [10.1016/j.jmarsys.2006.04.019](https://doi.org/10.1016/j.jmarsys.2006.04.019). URL: <https://www.sciencedirect.com/science/article/pii/S0924796307000668> (visited on 04/26/2024).
- Heywood, K., E. Barton, and J. Simpson (Jan. 1990). “The effects of flow disturbance by an oceanic island”. In: *Journal of Marine Research* 48.1. URL: https://elischolar.library.yale.edu/journal_of_marine_research/1961.

- Martinez, E. and K. Maamaatuaiahutapu (Sept. 2004). “Island mass effect in the Marquesas Islands: Time variation”. en. In: *Geophysical Research Letters* 31.18, 2004GL020682. ISSN: 0094-8276, 1944-8007. DOI: [10.1029/2004GL020682](https://doi.org/10.1029/2004GL020682). URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2004GL020682> (visited on 04/03/2024).
- Raapoto, H. et al. (2018). “Modeling the Wake of the Marquesas Archipelago”. en. In: *Journal of Geophysical Research: Oceans* 123.2. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/2017JC013285> pp. 1213–1228. ISSN: 2169-9291. DOI: [10.1002/2017JC013285](https://doi.org/10.1002/2017JC013285). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/2017JC013285> (visited on 04/05/2024).
- Raapoto, H. et al. (2019). “Role of Iron in the Marquesas Island Mass Effect”. en. In: *Journal of Geophysical Research: Oceans* 124.11. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/2019JC015275> pp. 7781–7796. ISSN: 2169-9291. DOI: [10.1029/2019JC015275](https://doi.org/10.1029/2019JC015275). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1029/2019JC015275> (visited on 04/03/2024).
- Signorini, S. R., C. R. McClain, and Y. Dandonneau (Oct. 1999). “Mixing and phytoplankton bloom in the wake of the Marquesas Islands”. en. In: *Geophysical Research Letters* 26.20, pp. 3121–3124. ISSN: 0094-8276, 1944-8007. DOI: [10.1029/1999GL010470](https://doi.org/10.1029/1999GL010470). URL: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/1999GL010470> (visited on 04/03/2024).
- Strutton, P. G. and F. P. Chavez (2000). “Primary productivity in the equatorial Pacific during the 1997–1998 El Niño”. en. In: *Journal of Geophysical Research: Oceans* 105.C11. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/1999JC000056>, pp. 26089–26101. ISSN: 2156-2202. DOI: [10.1029/1999JC000056](https://doi.org/10.1029/1999JC000056). URL: <https://onlinelibrary.wiley.com/doi/abs/10.1029/1999JC000056> (visited on 04/28/2024).
- Wong, A. P. S. et al. (Sept. 2020). “Argo Data 1999–2019: Two Million Temperature-Salinity Profiles and Subsurface Velocity Observations From a Global Array of Profiling Floats”. English. In: *Frontiers in Marine Science* 7. Publisher: Frontiers. ISSN: 2296-7745. DOI: [10.3389/fmars.2020.00700](https://doi.org/10.3389/fmars.2020.00700). URL: <https://www.frontiersin.org/articles/10.3389/fmars.2020.00700> (visited on 05/06/2024).

7 Appendix A: fluorometer calibration

The CTD fluorometer was calibrated using an chlorophyll extraction procedure. We collect 500 mL water from 45 Niskin bottles and use 0.45 μm filters to extract phytoplankton above this size threshold. The samples were then lysed with acetone and centrifuged, then analyzed with a pre-calibrated shipboard fluorometer. An additional step was taken to calibrate the RBR CTD from the carousel fluorometer calibration since the fluorometer voltage was different for the two CTDs. This was done by doing casts of the two CTDs during the same deployment. We employ a least squares linear fit to relate the two voltages. The standard curves are shown below.

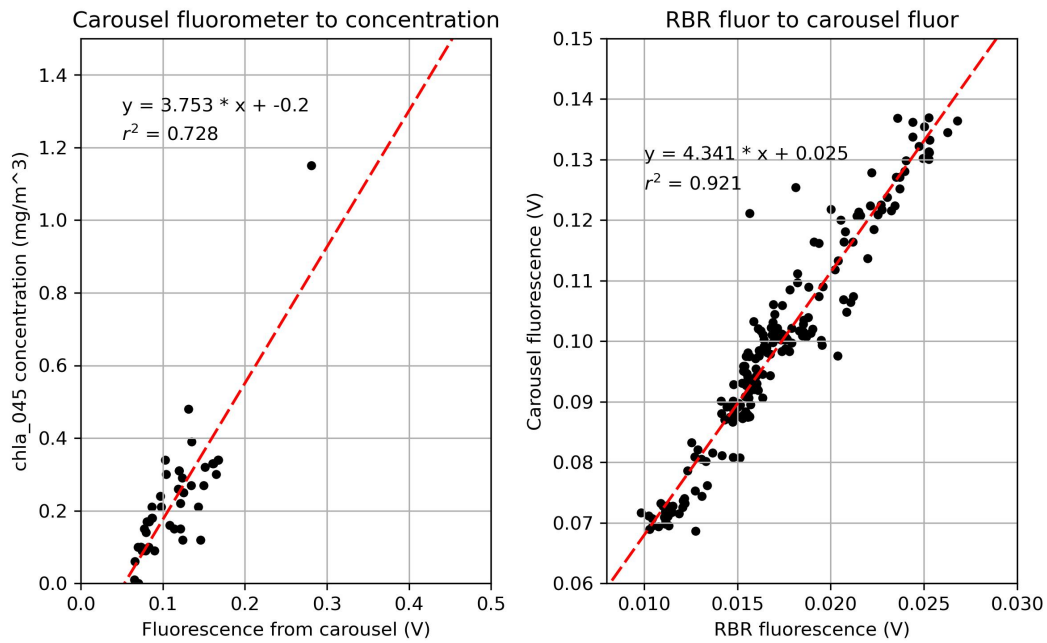


Figure S1: (a): carousel fluorometer to concentration; (b) RBR fluorometer voltage to carousel fluorometer voltage

8 Appendix B: complete tow data

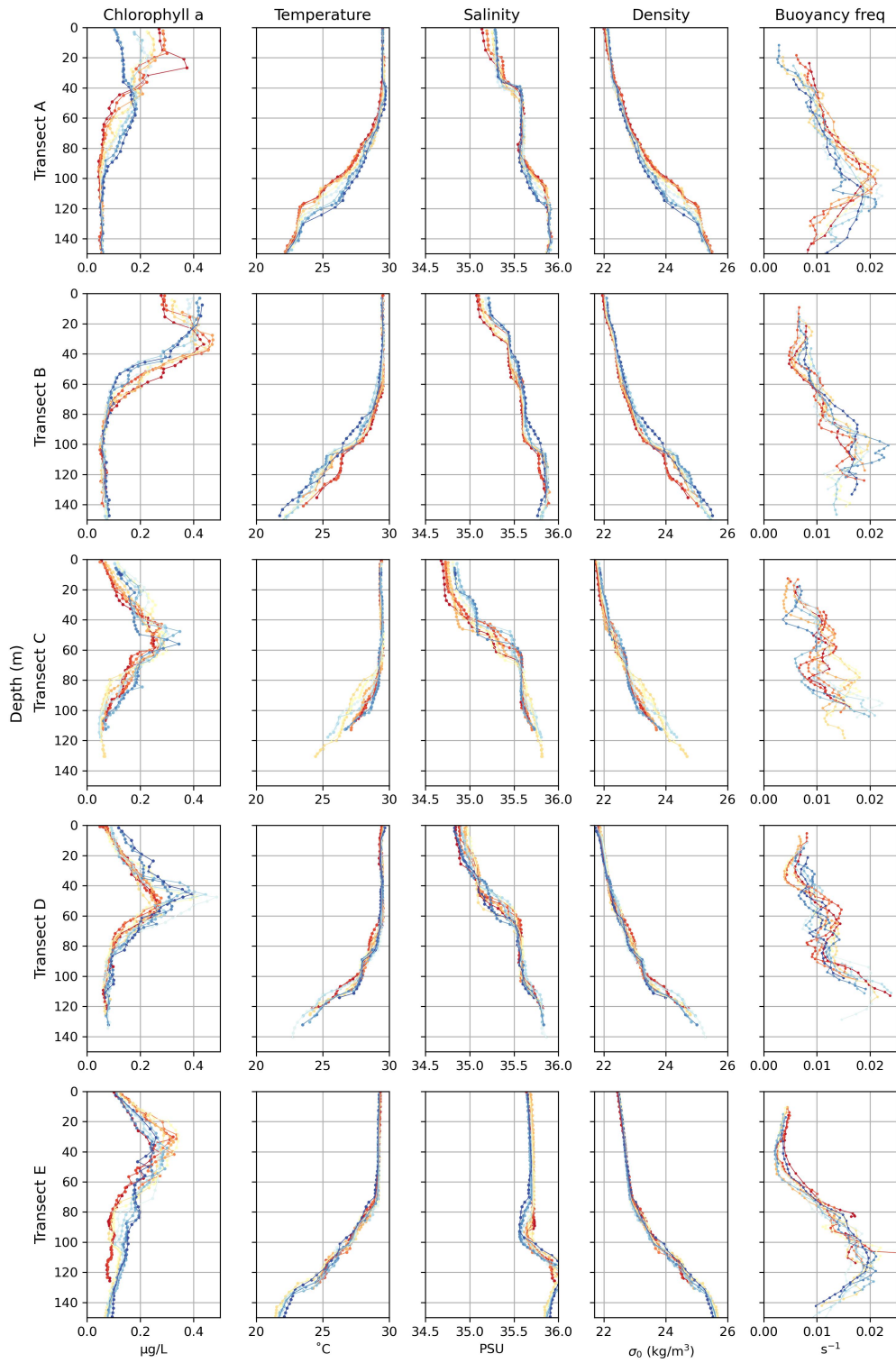


Figure S2: same as Figure 3, but with all transects.

9 Appendix C: additional timeseries figures

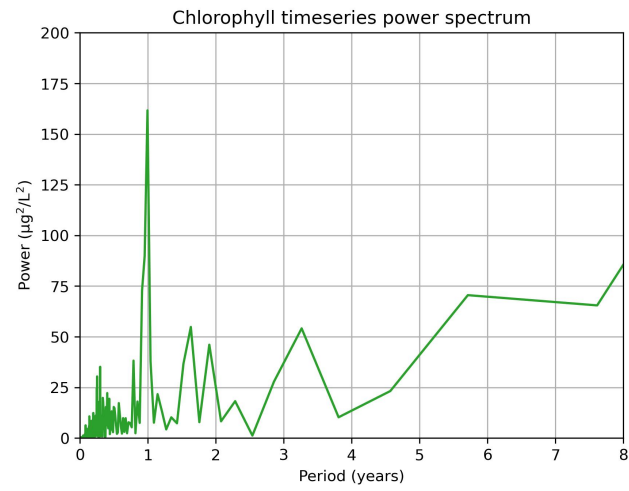


Figure S3: power spectrum of chlorophyll timeseries in Figure 5 calculated using Fast Fourier Transform.